

bwarea Function Test Results

Test Case 1: Empty Image

"BW:"

0. 0. 0. 0. 0.

0. 0. 0. 0. 0.

0. 0. 0. 0. 0.

0. 0. 0. 0. 0.

0. 0. 0. 0. 0.

"Sum of pixels:" 0

"Area:" 0



Test Case 2: Single Pixel Object

"BW:"

0. 0. 0. 0. 0.

0. 0. 0. 0. 0.

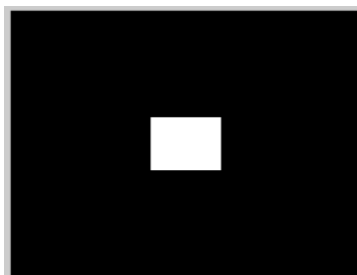
0. 0. 1. 0. 0.

0. 0. 0. 0. 0.

0. 0. 0. 0. 0.

"Sum of pixels:" 1

"Area:" 1



Test Case 3: Completely White Image

"BW:"

1. 1. 1. 1. 1.

1. 1. 1. 1. 1.

1. 1. 1. 1. 1.

1. 1. 1. 1. 1.

1. 1. 1. 1. 1.

"Sum of pixels:" 25

"Area:" 25



Test Case 4: Adjacent Pixels

"BW:"

0. 0. 0. 0. 0.

0. 0. 0. 0. 0.

0. 0. 1. 1. 0.

0. 0. 0. 0. 0.

0. 0. 0. 0. 0.

"Sum of pixels:" 2

"Area:" 2



Test Case 5: Diagonal Pixels

"BW:"

0. 0. 0. 0. 0.

0. 1. 0. 0. 0.

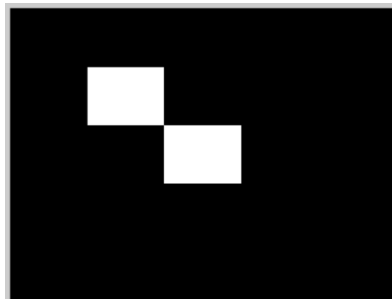
0. 0. 1. 0. 0.

0. 0. 0. 0. 0.

0. 0. 0. 0. 0.

"Sum of pixels:" 2

"Area:" 2.25



Test Case 6: Thin Horizontal Line

"BW:"

0. 0. 0. 0. 0. 0. 0.

0. 0. 0. 0. 0. 0. 0.

0. 0. 0. 0. 0. 0. 0.

0. 1. 1. 1. 1. 1. 0.

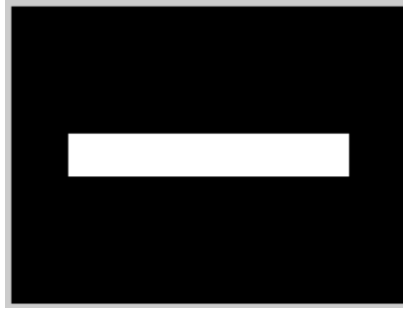
0. 0. 0. 0. 0. 0. 0.

0. 0. 0. 0. 0. 0. 0.

0. 0. 0. 0. 0. 0. 0.

"Sum of pixels:" 5

"Area:" 5



Test Case 7: Thin Vertical Line

"BW:"

```
0. 0. 0. 0. 0. 0. 0.
0. 0. 0. 1. 0. 0. 0.
0. 0. 0. 1. 0. 0. 0.
0. 0. 0. 1. 0. 0. 0.
0. 0. 0. 1. 0. 0. 0.
0. 0. 0. 1. 0. 0. 0.
0. 0. 0. 0. 0. 0. 0.
```

"Sum of pixels:" 5

"Area:" 5



Test Case 8: Multiple Objects

"BW:"

```
0. 0. 0. 0. 0. 0. 0.
0. 1. 1. 0. 0. 0. 0.
```

```
0. 0. 0. 0. 0. 0. 0. 0.
0. 0. 0. 0. 0. 0. 0. 0.
0. 0. 0. 0. 0. 0. 0. 0.
0. 0. 0. 0. 0. 1. 0. 0.
0. 0. 0. 0. 0. 1. 0. 0.
0. 0. 0. 0. 0. 0. 0. 0.
```

"Sum of pixels:" 4

"Area:" 4



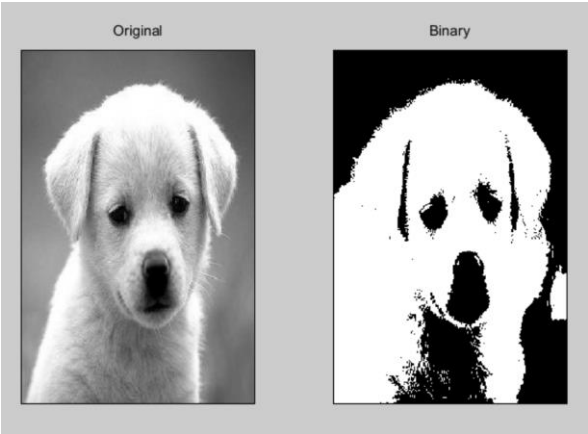
Test Case 9: Thresholded Binary Image

"BW (cropped view):"

```
F F F F F F F F F F
F F F F F F F F F F
F F F F F F F F F F
F F F F F F F F F F
F F F F F F F F F F
F F F F F F F F F F
F F F F F F F F F F
F F F F F F F F F F
F F F F F F F F F F
F F F F F F F F F F
```

"Sum of pixels:" 59530

"Area:" 59832.25



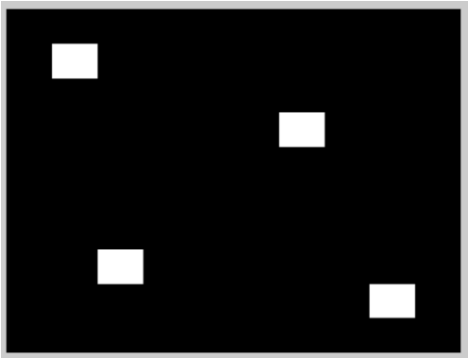
Test Case 10: Sparse Noise Pixels

"BW:"

```
0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0.
0. 1. 0. 0. 0. 0. 0. 0. 0. 0. 0.
0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0.
0. 0. 0. 0. 0. 0. 1. 0. 0. 0. 0.
0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0.
0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0.
0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0.
0. 0. 1. 0. 0. 0. 0. 0. 0. 0. 0.
0. 0. 0. 0. 0. 0. 0. 0. 1. 0. 0.
0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0.
```

"Sum of pixels:" 4

"Area:" 4



Test Case 11: Complex Shape

"BW:"

0. 0. 1. 1. 0. 0.

0. 1. 1. 1. 1. 0.

1. 1. 1. 1. 1. 1.

0. 1. 1. 1. 1. 0.

0. 0. 1. 1. 0. 0.

"Sum of pixels:" 18

"Area:" 19

